

Isaac Mubiru

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PROFESSIONAL SUMMARY

Backend engineer, **very proficient in Node.js and JavaScript**, with 5+ years building and operating complex production platforms. Led the end-to-end migration of **20+ applications** from Parse/MongoDB to Firebase/GCP with **zero downtime** reporting to the CTO, built the **real-time two-way MongoDB ↔ Firestore sync pipeline** on Google Pub/Sub that carried live traffic throughout, and architected a **microservices backend on Apache Kafka, Docker and Kubernetes**. Also built **DeFi backends on Web3**, where every call moves real value and cannot be undone. Now applying that same correctness-first discipline to secure, scalable wallet infrastructure. Open source: github.com/zac-09

CORE COMPETENCIES

Node.js & JavaScript (5+ yrs)

Microservices Architecture

Complex Platform Development

MongoDB & SQL

Distributed Systems (Kafka, NATS, gRPC)

Scalable Real-Time Backends

Zero-Downtime Migrations

Secure Backend Practices

WORK EXPERIENCE

MTailor

Sep 2022 – Present

Software Engineer (US Remote, reports to CTO)

- Built **real-time two-way sync between MongoDB and Firestore using Node.js and Google Pub/Sub** for message processing — the pipeline carried live production traffic in both directions throughout a months-long cutover.
- Led the end-to-end migration of **20+ production applications** from Parse/MongoDB to Firebase/GCP with **zero downtime**, reporting directly to the CTO.
- Saved the company **\$5,000/month** by migrating all services off AWS — consolidating a two-cloud estate onto one platform and its access model.
- Migrated file storage from Amazon S3 to Google Cloud Storage via a Python migration script.
- Implemented Express Shipping end to end, generating an additional **\$40 revenue per order**.
- Built a 3D visualisation feature using video overlay and ffmpeg, increasing buyer conversion.

Stack: Node.js, JavaScript, Firebase/Firestore, MongoDB, GCP Pub/Sub, AWS S3/EC2/EBS, Python, React

MTailor (Contractor)

Mar 2022 – Jul 2022

Backend Migration Contractor (US Remote)

- Hired via Upwork to execute a complex backend migration from Parse/**MongoDB** to Firebase Firestore and GCP.
- Built the migration script reading from MongoDB and writing to Firestore after complex processing logic.
- Wrote the **Node.js and Python SDK documentation** fellow engineers onboarded from, and trained the Ops team on the new dashboard and backup procedures — newcomer to teacher on an unfamiliar stack in a single engagement.

Stack: Node.js, JavaScript, MongoDB, Firebase, GCP, AWS

Dr Wealth

Aug 2021 – Jan 2022

Software Engineer (Singapore Remote)

- Built ad-hoc jobs querying over **2 million Firestore records** to keep customer-facing prices current from Morningstar APIs on a real-time PWA.
- Extended backend APIs using Firebase Cloud Functions and **Express.js** hosted on Heroku.
- Added responsive UI pages using React, Tailwind, HTML.

Stack: Node.js, JavaScript, Firebase, Express.js, React

Mind2matter

Jan 2022 – Jul 2022

Software Engineer (US Remote, reports to CTO)

- Built **backends for DeFi applications using Web3 and Node.js** — value-moving transaction flows on decentralized networks, where every call is irreversible.
- Delivered React UIs from Figma designs under tight client timelines.

Stack: Node.js, JavaScript, Web3, Firebase, React

CodeBits

Jan 2020 – Jul 2021

Software Engineer / Team Lead (Uganda Remote)

- Architected a **microservices backend** for FIDA Uganda's case management app using **Apache Kafka, Docker and Kubernetes**, with gRPC for synchronous service calls.
- Led a team of 4 developers building legal-tech systems for NGOs (FIDA Uganda, LASPNET Uganda).
- Built a USSD service for legal aid providers communicating with a Node.js backend via gRPC — web, mobile and USSD clients served off the same services.
- Developed a cross-platform mobile app (React Native + Expo, iOS & Android) with Firebase push notifications.

Stack: Node.js, JavaScript, Apache Kafka, NATS Streaming, gRPC, Docker, Kubernetes, Firebase

PROJECTS

Real-time Two-Way Sync Pipeline (MongoDB ↔ Firestore) live production traffic

Node.js consumers on Google Pub/Sub keeping two authoritative datastores consistent under concurrent production writes from both sides, across a months-long cutover. Designed so replays and out-of-order delivery were safe — at-least-once delivery treated as an idempotency problem by default.

Node.js, JavaScript, GCP Pub/Sub, MongoDB, Firestore, idempotent consumers

Parse → Firebase Platform Migration (MTailor) 20+ apps, zero downtime

End-to-end migration of 20+ production applications, owned from plan through cutover and rollback, reporting to the CTO. Per-app sequencing with a reversible step at every stage; \$5,000/month infrastructure saving on completion.

Node.js, MongoDB, Firestore, GCP, AWS S3/EC2/EBS, Python

Event-Driven Microservices Backend (FIDA Uganda) Kafka + Kubernetes

Architected service boundaries around Kafka events, deployed on Docker and Kubernetes, with gRPC for synchronous calls and NATS Streaming for lighter-weight messaging. Served web, mobile and USSD clients from one set of services.

Node.js, Apache Kafka, NATS Streaming, gRPC, Docker, Kubernetes

DeFi Application Backends (Web3) value-moving flows

Node.js backends for decentralized finance applications on Web3, built to agency deadlines. Every on-chain call treated as irreversible — validate before signing, make retries safe, keep the API contract stable.

Node.js, JavaScript, Web3, decentralized networks

EDUCATION

B.Sc. Computer Engineering — Makerere University

Kampala, Uganda

Aug 2018 – Sep 2022

CERTIFICATIONS

Open-source contributor & GitHub portfolio — github.com/zac-09

Ongoing

SKILLS

Proficient: Node.js, JavaScript, MongoDB, SQL, Firebase, GCP, React, Python, Git, Linux

Distributed & Infra: Apache Kafka, NATS Streaming, gRPC, Google Pub/Sub, Docker, Kubernetes, Express.js, AWS

Intermediate: Go, Rust, React Native, Web3

Languages: English (fluent, 5+ yrs US/Singapore remote), Kiswahili